

REVIEW

The prognostic value of p53 and Ki-67 expression status in penile cancer: a systematic review and meta-analysis



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ABSTRACT

This meta-analysis aimed to explore penile cancer survival based on p53 and Ki-67 status. A systematic literature search identified studies assessing overall and cancer-specific survival after penile cancer with p53 or Ki-67 expression. Pooled hazard ratios (HRs) and restricted mean survival time (RMST) differences were calculated using a random-effects model. The analysis included 930 and 391 men for p53 expression in overall survival (OS) and cancer-specific survival (CSS), respectively. Those with the p53 mutant pattern exhibited significantly worse OS and CSS compared to the p53 wild-type pattern (HR_{OS}=2.42, 95% CI 1.75–3.34; HR_{CSS}=4.18, 95% CI 1.87–9.35). The 5-year RMST difference for OS, according to p53 status, was –10.62 months (95% CI –16.20 to –5.03). We included 202 men with penile cancer tested for Ki-67 expression and found an HR of CSS of 1.96 (95% CI 1.15–3.32). In conclusion, men with the p53 mutant pattern penile cancer have worse OS and CSS than men with p53 wild-type pattern penile cancer. Evidence regarding Ki-67 status was sparse, but the pooled estimate indicated that penile cancers with high Ki-67 expression may have a slightly worse CSS than those with low Ki-67 expression. These findings may inform clinicians when planning the best management and follow-up strategy for penile cancer patients.

1. Introduction

Penile cancer is a rare malignancy, with approximately 36,000 new cases diagnosed and 13,000 deaths globally in 2020.¹ The incidence varies by geographical region, with a higher incidence in less developed countries.² Most penile cancers are squamous cell carcinomas (SCC), and the mean age at diagnosis is around 60–68 years.^{3–5} Penile carcinogenesis consists of two major aetiological pathways, one involving oncogenic human papillomavirus (HPV), and the other associated with non-viral mechanisms, including inflammatory conditions, smoking and lack of circumcision.^{6–8}

In studies of anogenital and oropharyngeal cancer, it has been found that tumours positive for HPV have superior survival compared to HPV negative.^{9–12} In addition, we have previously found that the detection of p16 expression by immunohistochemical (IHC) staining is a marker of prognostic value in patients with penile, vulva, vaginal and oropharyngeal cancer.^{9,11–13} In addition to p16, some other IHC markers seem to be of interest in the prediction of survival after penile cancer, including p53

and Ki-67 expression. Both p53 and Ki-67 are cell cycle-associated proteins. Ki-67 is a marker of cell proliferation and predicts malignant behaviour in some cancer types.^{14–17} The tumour suppressor protein p53 is often mutated in cancers, leading to an accumulation of the protein in malignant cells. In some anogenital SCC, it has been found that HPV E6 protein inactivates p53, causing downregulation of expression.¹⁸ Because of small study populations, existing studies examining survival after penile cancer according to p53 and Ki-67 expression status are often inconclusive. Identifying prognostic biomarkers in penile cancer, as in other cancers, would allow for better patient stratification based on the risk of cancer progression, ultimately guiding treatment decisions.

The aim of this study was to examine the association of p53 and Ki-67 expression status with overall survival (OS) and cancer-specific survival (CSS) in men diagnosed with penile cancer. In this systematic review and meta-analysis, we conducted a comprehensive literature search using multiple electronic databases to identify relevant studies to estimate OS, CSS, and restricted mean survival time (RMST) difference for men with penile cancer with p53 mutant pattern compared with p53 wild-type, and

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penile cancers with high Ki-67 expression compared with low Ki-67 expression.

2. Material and methods

We followed the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines,¹⁹ and the study protocol was registered at PROSPERO (CRD42022373058).

2.1. Search strategy and selection criteria

A systematic literature search of the electronic databases PubMed, Embase and Cochrane was conducted using a combination of search terms for penile cancer, p53, Ki-67 and survival. We used text words and Medical Subject Headings (PubMed, Cochrane Library) or Emtree headings (EMBASE); detailed search terms are provided in [Supplementary Table 1](#). Studies published between 1 January 2000 and 17 November 2023 were included. The search was restricted to English-language publications, and only peer-reviewed studies were included. The reference lists of the included studies were reviewed to identify other relevant studies. All titles, abstracts and full texts were reviewed independently by two authors (FLS and SL), and any inconsistencies in the selection process were discussed with a third reviewer (SKK) until a consensus was reached.

Studies were eligible for inclusion if they examined OS or CSS after penile cancer, comparing p53 mutant pattern or high Ki-67 expression with p53 wild-type pattern or low Ki-67 expression, respectively. Studies using IHC staining to detect p53 and/or Ki-67 expression from histologically verified samples of penile cancer were eligible for inclusion. In case of overlapping study populations, the study with the most comprehensive data was included.

2.2. Definition of p53 and Ki-67 status

Evaluation of p53 and Ki-67 status was based on IHC staining. As molecular TP53 status and IHC staining have shown close accordance, strong nuclear positivity was defined as a p53 mutant pattern, and heterogeneous nuclear staining was defined as a p53 wild-type pattern.^{20,21} Most studies defined p53-null tumours as mutant/positive, though earlier studies often excluded null tumours from mutant cases due to evolving definitions. Complete p53 negativity in tumour cells is now widely recognised as mutant status, but in some studies, tumours were only classified as mutant if a defined percentage of cells showed staining, thereby excluding null tumours. Ki-67 status was defined as either low or high expression based on the individual cut-off limits reported in the papers.

2.3. Data extraction

Two authors (FLS and SL) independently extracted the data, and inconsistencies were discussed with a third author (SKK) until a consensus was reached. We extracted the following variables from each study: first author, year of publication, year of sample collection, country, continent, age (mean, median, range), follow-up time, histology of the tumour, tumour stage and grade, treatment modality, tissue type, definition of p53 mutant and wild-type pattern and/or high and low Ki-67 expression, number of reviewers of IHC staining, sample size, number of p53 cases with wild-type/mutant pattern and/or low/high Ki-67 expression and survival endpoints.

2.4. Quality assessment

Based on guidelines from Reporting Recommendations for Tumor Marker Prognostic Studies (REMARK), we evaluated the quality of the included studies.^{22,23} The studies were assessed by two authors (FLS and SL) independently, and in case of inconsistencies, the studies were discussed with a third author (SKK). The studies were evaluated based on

predefined criteria ([Supplementary Table 2](#)) and scored between zero and seven, with a high score indicating a high study quality.

2.5. Statistical analysis

We used a random effects model to estimate the pooled hazard ratios (HRs) with corresponding 95% confidence intervals (CIs) of OS and CSS comparing penile cancer between cases exhibiting p53 mutant expression and p53 wild-type expression patterns. In the same way, we examined the CSS according to Ki-67 status. However, too few studies were available to calculate the pooled HR of OS for Ki-67 status.

In studies where Kaplan–Meier curves were available, individual patient data were reconstructed using the method described by Liu *et al.*²⁴ Based on the individual patient data, we estimated the HR and RMST difference. The HR was estimated by applying a Cox proportional hazard model, applying fixed-effects models for pooling. We estimated the 5-year HR for p53 studies and the 3-year HR for Ki-67 studies. In studies where the follow-up period was shorter than 5 or 3 years, respectively, the HR was estimated based on the available follow-up time. [Supplementary Table 3](#) presents an overview of the estimates extracted from the individual studies. The proportional hazards assumption was tested based on weighted Schoenfeld residuals, and the assumption was met. In studies where Kaplan–Meier curves were unavailable, the study-specific 5-year HRs were estimated using a previously described method by Moodie *et al.*²⁵ This method allowed us to perform meta-analysis, including studies with different lengths of follow-up time. The results from the pooled analyses are presented graphically with forest plots, including confidence intervals.

The RMST is an alternative method for comparison across groups. Individual patient data from studies presenting Kaplan–Meier curves were used to calculate RMST.²⁴ For p53 expression status, we estimated the overall RMST difference after 3 and 5 years of follow-up and the cancer-specific RMST difference after 5 years of follow-up. For Ki-67 expression status, the cancer-specific RMST difference after 2 years of follow-up was estimated.

Statistical heterogeneity across the studies was evaluated by the I^2 statistic, which describes the percentage of variation between studies.²⁶ A value of 0% indicates no observed heterogeneity, whereas values of 25, 50, and 75% correspond to low, moderate and high levels of heterogeneity, respectively. In addition, the significance of the heterogeneity across the studies was determined using the Cochran Q test, with a p value <0.05 considered statistically significant.²⁷ Publication bias was assessed using funnel plots and the Egger regression test.

All analyses were conducted using the statistical software R version 4.2.1,²⁸ with the packages IPDfromKM,²⁹ SurvRM2³⁰ and meta.³¹

3. Results

3.1. Search results

[Fig. 1](#) shows the flowchart of the study selection process. We identified 264 records in the electronic databases and after removing duplicates, 219 unique records were screened for potential relevance. We excluded 127 and 38 studies based on titles and abstracts, respectively. Thus, 54 studies were included for review of the full text. After a review of full texts, 34 studies were excluded, leaving 20 studies eligible for inclusion in the meta-analyses. Among the included studies, 15 reported survival according to p53 status,^{32–46} and seven examined survival according to Ki-67 status.^{32,45,47–51} Studies excluded due to reporting of other survival estimates than OS and CSS are presented in [Supplementary Table 4](#) ($n=3$).^{52–54} Two of the included studies had partly overlapping study populations.^{33,38} They included penile samples from the same hospital from 1980 until 1992. However, samples collected during a period of 39 years (1953–1980 and 1992–2009) were not overlapping, and therefore both studies were included in the meta-analysis. No studies were identified upon review of reference lists.

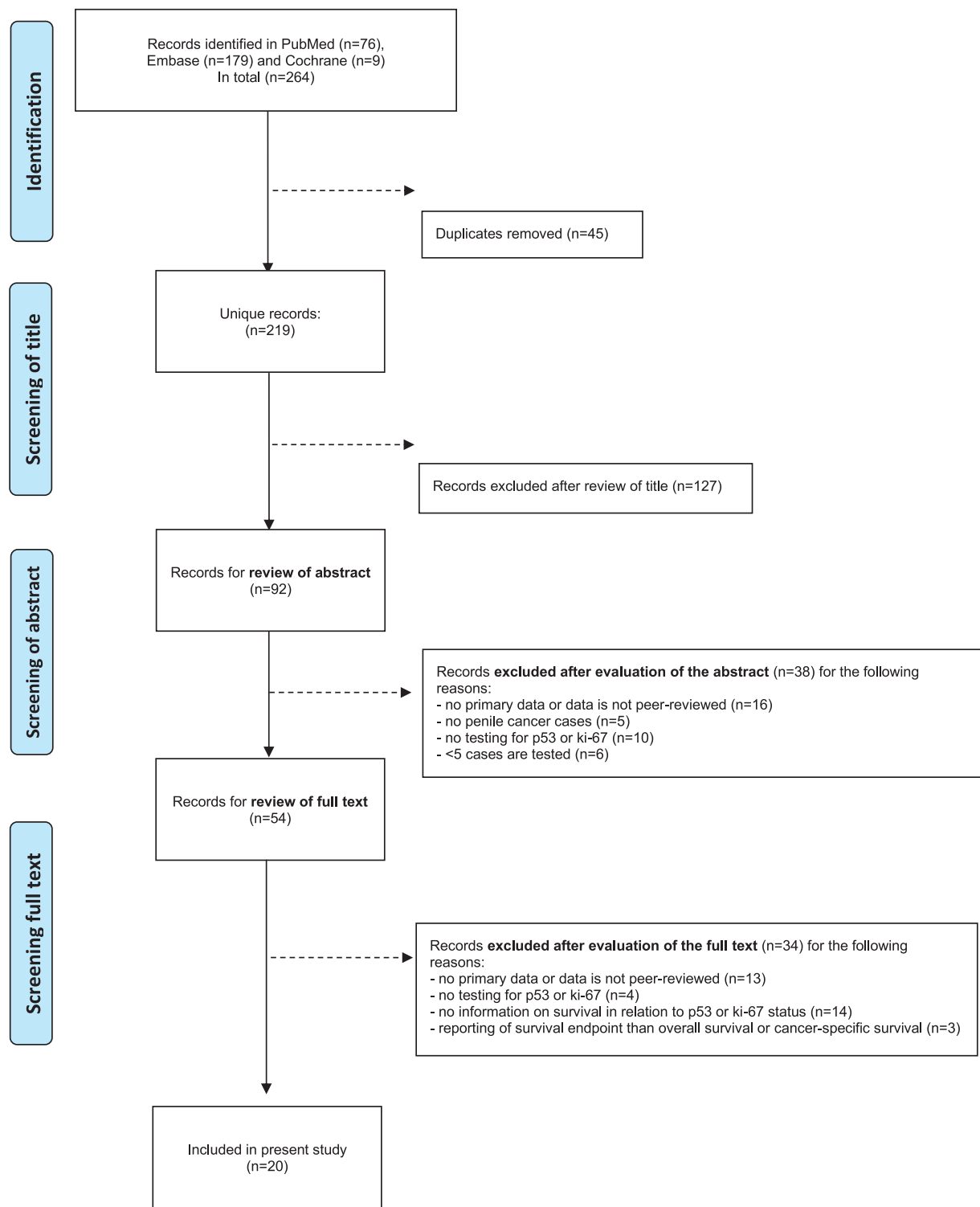


Fig. 1. Flowchart of the study selection process. The literature search was conducted on 17 November 2023.

3.2. Survival according to p53 expression status

3.2.1. Study characteristics

Characteristics and quality assessment of the studies examining survival after penile cancer according to p53 status are presented in Table 1. The studies were published between 2002 and 2023, all were retrospective observational studies, and the penile cancer samples were collected between 1953 and 2022. Six studies originated from Europe,^{35,40–42,44,46} four from South America,^{33,37,38,43} and four from

Asia.^{32,34,36,39} One study came from North America.⁴⁵ The mean or median age of the men with penile cancer ranged from 51 years to 70 years. The majority of studies (n=14) included only penile SCC histology,^{32,34–46} and one study included various histological types, of which 64% were SCC.³³ Most studies included tumours of stages 1–4 (n=7)^{32,36,37,39,42,43,46} or 1–3 (n=5).^{35,40,41,44,45} For p53 IHC testing, the majority of studies used fixed tissue (n=13).^{32,34–38,40–46} The definition of the p53 mutant pattern varied between studies with the study-specific IHC cut-off limits described in Table 1. The number of men with penile

Table 1

Study characteristics for studies reporting overall or cancer-specific survival after penile cancer according to p53 status

Author, year, reference	Year of sample collection	Country	Age (range), years	Histology	Tumour grade	Tumour stage	Treatment	Immunohistochemical nuclear staining cut-off limits: definition of expression status		Men with penile cancer, N	p53 status			Survival outcomes reported	Quality assessment score ^a
								p53 mutant pattern	p53 wild-type pattern		No. mutant	No. wild-type	% mutant		
Gil <i>et al.</i> , 2002 ⁴³	1979–1995	Brazil	Mean 57 (23–80)	SCC	G1–3	1–4 ^b	Surgery	>75% of tumour cells	<25% of tumour cells	11 ^c	4	7	25%	OS	2
Lopes <i>et al.</i> , 2002 ^{38,8}	1953–1992	Brazil	Mean 52 (27–77)	SCC	G1–2	2–4	Surgery	>20% of tumour cells	<20% of tumour cells	82	34	48	41%	OS	7
Martins <i>et al.</i> , 2002 ³⁷	1978–1995	Brazil	Median 59 ^d	SCC	G1–3 ^d	1–4 ^d	Mixed ^e	>10% of tumour cells	<10% of tumour cells	36	14	22	39%	CSS	6
Zhu <i>et al.</i> , 2007 ³²	1990–2005	China	Median 55 (27–75)	SCC	G1–4 ^f	1–4 ^f	Surgery	>20% of tumour cells	<20% of tumour cells	73	24	49	33%	CSS	5
Guerrero <i>et al.</i> , 2008 ⁴²	1988–2007	Spain	Mean 68 (28–81)	SCC	G1–3	1–4	Surgery	>20% of tumour cells	<20% of tumour cells	24	11	13	46%	OS	4
Bethune <i>et al.</i> , 2012 ⁴⁵	1997–2009	Canada	Mean 63 (36–95)	SCC	G1–3	1–3	Mixed ^e	>5% of tumour cells	<5% of tumour cells	43	17	26	40%	OS, CSS	5
Gunia <i>et al.</i> , 2012 ⁴¹	1993–2010	Germany	Mean 68	SCC	G1–3	1–3	Surgery	At least moderately strong expression in basal and suprabasal cells	Absent or weak expression restricted to basal cells	92	30	62	33%	CSS	6
Rocha <i>et al.</i> , 2012 ^{33,8}	1980–2009	Brazil	NA	Various ^h	G1–3	NA	NA	NA	NA	297	NA	NA	NA	OS, CSS	2
Ferrandiz-Pulido <i>et al.</i> , 2013 ⁴⁴	1987–2010	Spain	Median 70 (40–96)	SCC	G1–4	1–3	Surgery	>5% of tumour cells	<5% of tumour cells	67	32	35	48%	CSS	6
Liu <i>et al.</i> , 2013 ³⁹	1998–2011	China	Mean 51 (24–82)	SCC	G1–3	1–4	Mixed ^e	>20% of tumour cells	<20% of tumour cells	146	56	90	38%	OS	4
Kashofer <i>et al.</i> , 2017 ⁴⁰	NA	Austria	Mean 60 (41–85)	SCC	G1–3	1–3	Surgery	Continuous strong staining of basal keratinocyte nuclei	None or only individual basal keratinocyte showed a reaction	26	15	11	58%	OS	2
Prapiska <i>et al.</i> , 2019 ³⁴	2018	Indonesia	Mean 51	SCC	NA	2–4	Mixed ^e	>60% of tumour cells	<60% of tumour cells	33	13	20	39%	OS	2
Mohanty <i>et al.</i> , 2021 ³⁶	2009–2017	India	Median 65 (49–80)	SCC	G1–3	1–4	Mixed ^e	≥75% of tumour cells	<75% of tumour cells	123	70	53	57%	OS, CSS	7
Panic <i>et al.</i> , 2021 ³⁵	2009–2018	Germany	Mean 63 (39–90) ^d	SCC	G1–4 ^d	1–3 ^d	Mixed ^e	≥80% of tumour cells	<80% of tumour cells ⁱ	36	10	26	28%	OS	4
Hrudka <i>et al.</i> , 2023 ⁴⁶	2002–2022	Czech Republic	Mean 66 (31–92)	SCC	G1–3	1–4	Mixed ^e	Diffuse strong nuclear staining	Heterogeneous nuclear staining	152	27	125	18%	OS, CSS	5

CSS, cancer-specific survival; NA, not available; OS, overall survival; SCC, squamous cell carcinoma.

^a The studies were scored between 0 and 7, with a high score indicating a high study quality.^b The study used the criteria by Jackson to analyse the clinical stage of the penile cancers.^c Tumours with grade 2 and 3 immunohistochemical nuclear staining are not included.^d The information is for the overall study population and not exclusively for the penile cancer cases included in the survival analysis.^e Mixed treatment includes surgery and/or radiotherapy and/or chemotherapy.^f Tumour grade is reported as G1 and ≥G2 and tumour stage is reported as T1 and ≥T2.^g Studies by Lopes *et al.*³⁸ and Rocha *et al.*³³ have partly overlapping study populations.^h Among 297 penile cancers, a total of 189 were squamous cell carcinomas.ⁱ Tumours with loss of p53 expression were not included.

cancer included in the studies ranged from 11 to 297, and the prevalence of the p53 mutant pattern varied between 18% and 58%. The studies scored between two and seven points in the quality assessment, with half of the studies scoring five points or above.^{32,36–38,41,44–46}

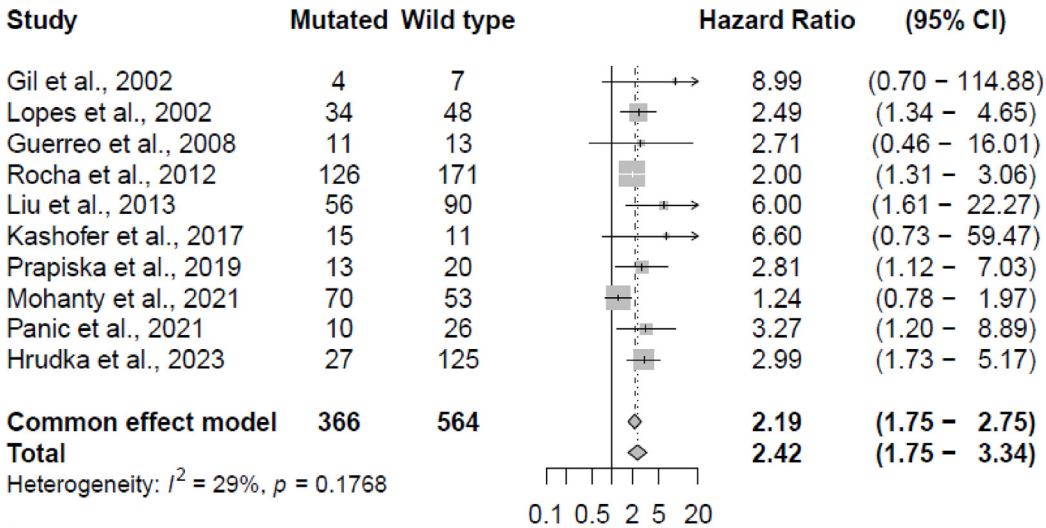
3.2.2. Overall survival

Fig. 2A presents the study-specific and pooled unadjusted HR of OS for men with penile cancers with a p53 mutant pattern compared with a p53 wild-type pattern. Ten studies^{33–36,38–40,42,43,46} reported information to be included in the meta-analysis, comprising a total of 930 men with penile cancer. All study-specific HRs were above 1, ranging from 1.24 to 8.99. Men with penile cancers with the p53 mutant pattern had significantly worse survival compared to those with the p53 wild-type pattern with a pooled HR of 2.42 (95% CI 1.75–3.34). The across-study heterogeneity was moderate ($I^2=29\%$). The funnel plot and Egger regression test indicated no strong evidence of publication bias (Supplementary Table 5). We performed two sensitivity analyses. In the first sensitivity analysis, we excluded Gil *et al.* because an artificial cause of death among penile cancers with wild-type p53 pattern was assumed in the main analysis, and Liu *et al.* as they did not report an HR at a specific follow-up time, but only the mean HR of OS during the entire follow-up period (median follow-up 42 months, range 13–108 months)^{39,43} and the results were similar (HR=2.22, 95% CI 1.62–3.04) (Supplementary Table 6). In the second sensitivity analysis, we only included the four studies where p53 mutant pattern was defined as strong staining, $\geq 75\%$ staining or complete absence,^{35,36,40,46} and the results were similar to the overall analyses

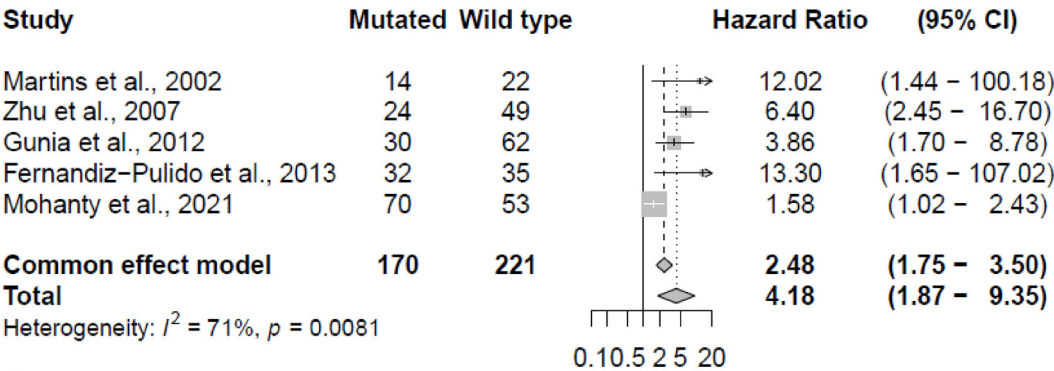
(HR=2.33, 95% CI 1.23–4.38) (Supplementary Table 7). Bethune *et al.* only reported a *p* value for the association between p53 status and OS and therefore could not be included in the pooled analysis. Based on 43 men with penile cancer, they reported that p53 status was not a predictor of OS.⁴⁵

The three largest studies, including respectively 297, 152 and 146 men with penile cancer,^{33,39,46} conducted multivariate analysis of OS for p53 mutant tumours compared with cancers with p53 wild-type patterns, taking into account other prognostic factors for survival, including e.g. tumour stage, grade, lympho-vascular invasion and budding (Supplementary Table 8). They found that men with tumours with mutant p53 patterns had significantly worse survival compared with cancers with p53 wild-type pattern, as Rocha *et al.* reported a HR of 1.83 (95% CI 1.11–3.04), Hrudka *et al.* 2.0 (95% CI 1.09–4.35), and Liu *et al.* found a HR of 6.0 (95% CI 1.62–22.28).^{33,39,46}

We also estimated the RMST difference after 3- and 5-years of follow-up comparing OS of men with p53 mutant pattern penile cancer to those with p53 wild-type pattern based on the studies where Kaplan–Meier curves were available^{33–36,38,46} (Supplementary Table 9). The 3-year RMST ranged from 18.0 to 29.8 months for men with the p53 mutant pattern, and 26.45 to 31.81 months for the p53 wild-type pattern. Thus, the difference in RMST at 3 years for the p53 mutant pattern compared with the p53 wild-type pattern was –6.00 months (95% CI –9.32 to –2.68). The 5-year RMST ranged from 24.6 to 45 months and 43.3 to 51 months for men with penile cancers with p53 mutant pattern and p53 wild-type pattern, respectively, with a 5-year RMST difference of –10.62 months (95% CI –16.20 to –5.03).



A



B

Fig. 2. Forest plot of (A) overall survival and (B) cancer-specific survival in men with penile cancers with p53 mutant pattern compared with p53 wild-type pattern.

3.2.3. Cancer-specific survival

Fig. 2B shows the study-specific and pooled unadjusted HR of CSS for p53 men with penile cancers with a p53 mutant pattern compared with p53 wild-type pattern. Five studies were included in the meta-analysis, comprising a total of 391 men with penile cancer, of which 170 and 221 cases showed p53 mutant and wild-type pattern, respectively.^{32,36,37,41,44} All study-specific HRs were above one, ranging from 1.58 to 13.30. The pooled 5-year HR of CSS was 4.18 (95% CI 1.87–9.35). The heterogeneity observed across studies was moderate ($I^2=71\%$) and no strong evidence of publication bias was found (Supplementary Table 5B). In a sensitivity analysis excluding one study where an artificial cause of death among men with a p53 wild-type pattern was assumed, we found no substantial change in the pooled HR (Supplementary Table 10).⁴⁴ Two studies only reported a p value for the association between CSS and p53 status among men with penile cancer.^{33,45} In one study by Rocha *et al.*, they included 297 men with penile cancer. They found that men with p53 wild-type pattern had a significantly better CSS than men with p53 mutant pattern ($p=0.003$),³³ and another smaller study from Canada, including 43 men with penile cancer, reported no association between CSS and p53 status.⁴⁵ Hrudka *et al.*⁴⁶ also found that men with p53 wild-type had a significantly better CSS (HR=3.33; 95% CI 1.47–7.69) than men with a p53 mutated pattern. However, Hrudka *et al.*⁴⁶ could not be included in the pooled estimate due to missing information on the CSS for p53-mutated and wild-type men.

Three smaller studies, including between 36 and 92 penile cancer cases, performed multivariable analyses and found a significant association between p53 expression status and CSS after adjusting for e.g. stage and grade of the tumour ($p=0.02$, 0.011, and 0.041).^{32,37,41} Gunia *et al.* included 92 men with penile SCC, of which 62 showed a p53 mutant pattern.⁴¹ They reported a HR of 3.20 (95% CI 1.05–9.76) after adjusting for tumour stage, grade, lymphovascular invasion and corpus spongiosum invasion and corpus cavernosa invasion.⁴¹ An overview of studies conducting multivariate analysis of CSS survival after penile cancer according to p53 status is provided in Supplementary Table 8.

We estimated the 5-year RMST difference comparing CSS in men with penile cancers with the p53 mutant pattern to those with the p53 wild-type pattern. The RMST difference was –12.44 months (95% CI –19.41 to –5.47) after 5 years of follow-up. However, this was based on only two studies where Kaplan–Meier curves were available.^{36,37}

3.3. Survival according to Ki-67 expression status

3.3.1. Study characteristics

Table 2 presents an overview of study characteristics and quality assessment of the seven studies examining OS or CSS according to Ki-67 expression status after penile cancer. The studies were published between 2007 and 2013. Most of the studies originated from Europe ($n=3$),^{47,48,50} followed by Asia ($n=2$).^{32,49} All studies were retrospective observational studies. The samples were collected between 1953 and 2011, and the histology of the penile cancers was SCC. The mean or median age of the men with penile cancer ranged from 56 to 69 years. The majority of studies included cancers of grade 1–3 ($n=5$).^{45,47,48,50,51} Four studies comprised cancers of stage 1–4.^{32,47,48,51} All studies used fixed tissue for Ki-67 IHC testing. The definition of high and low Ki-67 staining varied between studies (study-specific cut-off limits are described in Table 2), reflecting the varying Ki-67 prevalence between studies. The size of the study population ranged from 28 to 148 men with penile cancer. Most studies scored five in the quality assessment ($n=5$).^{32,45,47,48,50}

3.3.2. Overall survival

Supplementary Table 11 presents an overview of the results reported in studies examining OS according to Ki-67 status. Five studies examined OS after penile cancer according to Ki-67 status.^{45,47,49–51} Two of the largest studies by Guimaraes *et al.* ($n=125$) and Stankiewicz *et al.*

Table 2 Study characteristics for studies reporting survival after penile cancer according to Ki-67 status

Author, year, reference	Year of sample collection	Country	Age (range), years	Histology	Tumour grade	Tumour stage	Treatment	Immunohistochemical nuclear staining cut off limits: definition of expression status		Men with penile cancer, N	Ki-67 status		Survival outcomes reported	Quality assessment score ^a
								High Ki-67 expression	Low Ki-67 expression		No. high	No. low		
Guimaraes <i>et al.</i> , 2007 ⁵¹	1953–1997	Brazil	NA	SCC	G1–3	1–4	Surgery	>10% of tumour cells	<10% of tumour cells	125	62	63	OS	6
Protzel <i>et al.</i> , 2007 ⁴⁸	1993–2003	Germany	Median 69 (35–89)	SCC	G1–3	1–4	Mixed ^b	>50% of tumour cells	<50% of tumour cells	28	11	17	CSS	5
Zhu <i>et al.</i> , 2007 ³²	1990–2005	China	Median 55 (27–75)	SCC	G1–4 ^c	1–4 ^c	Surgery	>20% of tumour cells	<20% of tumour cells	73	26	47	CSS	5
Kayes <i>et al.</i> , 2009 ⁵⁰	1988–2007	England	Mean 62 ^d	SCC	G1–3	1–3 ^d	NA	NA	NA	118	NA	NA	OS	5
Bethune <i>et al.</i> , 2012 ⁴⁵	1997–2009	Canada	Mean 63 (36–95)	SCC	G1–3	1–3	Mixed ^b	>10% of tumour cells	<10% of tumour cells	43	36	7	OS, CSS	5
Stankiewicz <i>et al.</i> , 2012 ⁴⁷	2001–2007	England	NA	SCC	G1–3	1–4	NA	>40% of tumour cells	<40% of tumour cells	148 ^e	57	91	OS, CSS	5
Li <i>et al.</i> , 2013 ³⁹	2004–2011	China	Mean 56	SCC	NA	NA	NA	> the median value	< the median value	54	26	28	OS	2
			Median 61 (22–86)					of stained tumour cells ^f						

CSM, cancer specific mortality; CSS, cancer specific survival; DFS, disease free survival; IHC, immunohistochemistry; NA, not available; OS, overall survival; PFS, progression free survival; SCC, squamous cell carcinoma.

^a The studies were scored between 0 and 7, with a high score indicating a high study quality.

^b Mixed treatment includes surgery and/or radiotherapy and/or chemotherapy.

^c Zhu *et al.*³² reported tumour grade as G1 and $\geq$ G2 and tumour stage is reported as T1 and $\geq$ T2. Cocks *et al.* reported tumour stage as T1 and $\geq$ T1.

^d The information is for the overall study population including men with penile cancer who are not included in the survival analysis.

^e Follow-up data were available for 134 men with penile cancer.

^f Median value is unknown.

($n=148$) found that Ki-67 status was not significantly associated with OS ($p>0.05$ and $p=0.5538$).^{47,51} Similar results were reported in two smaller studies by Bethune *et al.* ($p>0.05$) and Li *et al.* ($RR=1.182$; 95% CI 0.40–3.47; $p=0.761$).^{45,49} A study from England including 118 men with penile SCC found that Ki-67 status was a predictor for OS in univariate analysis ($HR=1.023$; 95% CI 1.00–1.05); however, when adjusting for other prognostic factors, including age, stage, nodal status and vascular invasion, Ki-67 status was not an independent predictor for OS ($p>0.05$).⁵⁰

3.3.3. Cancer-specific survival

Four studies examined CSS according to Ki-67 status among men with penile cancer,^{32,45,47,48} of which three studies were included in the pooled analysis, comprising a total of 202 penile cancers ($N_{\text{high Ki-67 expression}}=76$, $N_{\text{low Ki-67 expression}}=126$).^{32,47,48} Men with high Ki-67 expression tumours had a slightly statistically non-significant worse CSS compared with those with low Ki-67 expression with a pooled HR of 1.96 (95% CI 1.15–3.32) (Fig. 3). The study heterogeneity was low ($I^2=0\%$). Bethune *et al.* could not be included in the pooled analysis, as they only provided a p value for the association between Ki-67 status and CSS.⁴⁵ They had 43 men with penile SCC and reported no significant association between Ki-67 status and CSS ($p>0.05$).⁴⁵ One study conducted multivariate analysis adjusting for age, stage, grade, lymphatic and vascular embolisation, E-cadherin status, and MMP-9 status and found that Ki-67 status was not a prognostic factor for CSS ($p=0.323$).³² The funnel plot showed no strong publication bias (Supplementary Table 5c).

Finally, we estimated the 2-year RMST difference for penile cancers with high Ki-67 expression compared with low Ki-67 expression. Based on two studies where Kaplan–Meier curves were provided, the RMST difference was -3.44 months (95% CI -7.97 to 1.10).^{47,48}

4. Discussion

To our knowledge, this is the first meta-analysis examining survival after penile cancer according to p53 and Ki-67 expression status. We found that men with penile cancer with a p53 mutant pattern had significantly worse OS and CSS compared with p53 wild-type tumours. In addition, studies adjusting for other prognostic factors, including stage, grade and lympho-vascular invasion, concluded that p53 status could be an independent prognostic factor for survival after penile cancer. We found that 5 years after the penile cancer diagnosis, men with a p53 wild-type pattern, on average, had lived approximately 11 months longer than men with a p53 mutant pattern. In contrast, the included studies reported no association between Ki-67 status and OS but seem to suggest that men with penile cancers with high Ki-67 expression might have a slightly worse CSS compared with cancers with low Ki-67 expression. However, this result was based on relatively few studies.

The p53 expression status as a prognosticator for survival has been suggested in other anogenital and head and neck cancers.^{9,13,55,56} We have previously reported that women with p53 wild-type vulvar tumours have superior survival compared with p53 mutant tumours.¹³ Studies have also suggested that p53 expression has prognostic value in vaginal

cancer.⁹ The possible prognostic value of Ki-67 expression status in malignancies differs between cancer sites.^{57,58} Ki-67 expression has been shown to be a reliable marker for survival in patients with e.g. breast, tongue, cervical, prostate, renal and neuroendocrine cancer.^{14,15,17,59–62} Our results indicated that Ki-67 expression status might be associated with CSS in men diagnosed with penile cancer. However, the results were limited by few and smaller studies, and therefore, larger studies are still warranted.

A recent review examined p53 and Ki-67 status in predicting lymph node metastasis and CSS in men diagnosed with penile cancer and concluded that p53 expression has prognostic value for CSS, but more studies are needed for Ki-67 status.⁶³ The present study is the first meta-analysis to calculate a pooled estimate for OS and CSS for penile cancers with a p53 mutant pattern compared with p53 wild-type pattern cancers, and CSS for high Ki-67 expression compared with low Ki-67 expression. In addition, we also examined RMST differences according to p53 and Ki-67 expression status detected by IHC staining. It is a strength of the study that two authors independently assessed titles, abstracts and full texts and extracted the data after conducting a thorough literature search of relevant studies in several electronic databases. The studies examining OS or CSS survival after penile cancer according to p53 were generally limited by small study population. However, by combining the studies, we were able to include 930 and 391 men with penile cancer tested for p53 expression and examined OS and CSS after cancer diagnosis, respectively. In addition, by extracting individual patient data from Kaplan–Meier curves, we performed a standard survival analysis and were able to get estimates that were based on more information and thereby considered more accurate than estimates obtained using the method described by Moodie *et al.*²⁵

Some limitations should also be considered. Due to the available data, we were not able to take into account other prognostic factors such as the stage of the tumours in the analyses. Therefore, it remains uncertain to what extent p53 status alone influences the survival after penile cancer. The definition of p53 mutant/wild-type expression pattern and high/low Ki-67 expression varied between studies, which we were unable to consider in the analyses. The definition of p53 mutant is currently more often characterised as strong nuclear staining or total absence of nuclear expression, while the definition of wild-type is heterogeneous nuclear staining.²¹ We did a sensitivity analysis using only four of the most recent studies that used a similar criterion for p53 status and found a similar result. Finally, the survival analysis according to Ki-67 status was based on only a few studies.

Survival according to p53 and Ki-67 expression status is of clinical interest, as testing for p53 expression in penile tumour tissue could help in planning the management and follow-up strategy. Our results indicate that men with penile cancers with p53 mutant patterns might need to be followed more closely compared with men with a p53 wild-type pattern. It has been suggested that combined testing for multiple markers, including HPV, p16 and p53, increases the accuracy in predicting patient survival compared with individual markers alone. In head and neck cancer, the tumour is tested for a group of markers that influence the treatment regime and follow-up program offered to the patients.⁵⁶

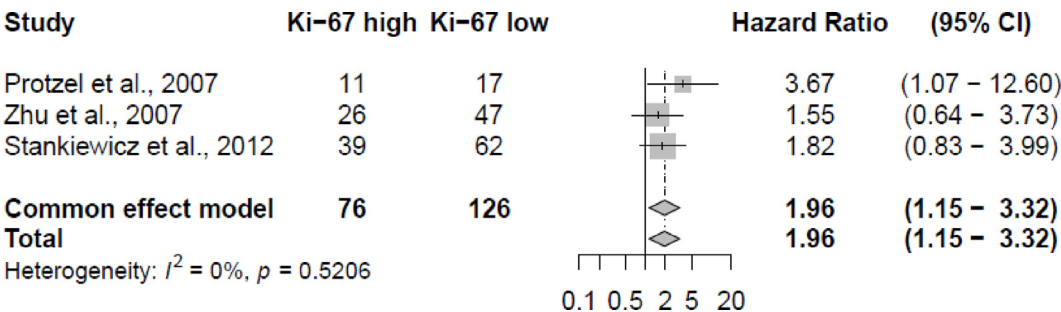


Fig. 3. Forest plot and funnel plot of cancer-specific survival in men with penile cancers with positive Ki-67 expression compared with negative Ki-67 expression.

However, studies on combined testing of markers are still needed for penile cancer.

5. Conclusion

In conclusion, in this systematic review and meta-analysis, we found that p53 status in penile cancer had a significant prognostic value for both OS and CSS. This was also apparent in studies taking into account known prognostic markers. Few studies with smaller study populations limited the interpretation of the results for Ki-67. However, it indicated that men with high Ki-67 expression might have a worse CSS compared to those with low Ki-67 expression. These findings suggest that testing for p53 and possibly Ki-67 expression may be useful when planning the management and follow-up strategy of men with penile cancer.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of interest and sources of funding

The authors state that there are no conflicts of interest to disclose. SKK has received research grants through her institution from Merck, not related to the present study.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pathol.2024.11.006>.

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